Supplemental Material for "Defect-aware graph transformer for impurity incorporation energies and axis-resolved chemical transfer in two-dimensional materials"

Yiming Hua[1]

[1] *Wuhan University, Wuhan, China*

(Dated: August 9, 2026)

I. REPRODUCIBILITY SPECIFICATION

Table I records the fixed model and optimization settings used in the formal comparison. The resolved configuration for every run, including split and seed, is archived under `configs/prm/`; the corresponding run manifests bind those configurations to checkpoints, predictions, environments, and output hashes.

TABLE I. Fixed settings for the promoted DART model and the prespecified periodic SchNet comparator. Settings not shown as different were shared.

Setting	DART (`g111`)	Periodic SchNet (additive readout)
Atomic input	Fixed 128-dimensional ct-UAE vector, nine tabulated elemental attributes, and a learned unique-impurity flag	Learned atomic-number embedding
Interaction stack	Three 5-Å local layers with 32 radial and 32 angular functions, followed by two four-head geometric Transformer blocks with 12-Å radial bias	Six 5-Å interaction blocks with 50 Gaussian functions
Hidden width and readout	128 channels; gated fusion of an atom-attention sum and channelwise maximum	128 channels and filters; atomwise additive scalar readout
Trainable parameters	820,558	455,937
Optimization	AdamW, batch size 64	AdamW, batch size 32
Shared schedule	150 epochs; MAE objective; learning rate 5×10^{-5} to 5×10^{-4} over a ten-epoch linear warmup, then cosine decay to 10^{-6}; weight decay 10^{-4}; gradient clipping at 5.	
Shared sampling and regularization	Training-target standardization; host-balanced sampling of 200 structures per host per epoch; 0.03-eV Gaussian target noise; no coordinate or strain augmentation. DART additionally uses 0.1 dropout.	

Factorial assignment seeds were 42–46 with paired model seeds 142–146. Transfer runs used DART seeds 242–244 and SchNet seeds 342–344, with one prespecified seed per fold for random and host–impurity-pair cross-validation and three seeds per split for host, impurity, and chemistry-block evaluation. The dedicated DART uncertainty ensemble used seeds 442–446.

The descriptor search was also fixed before test aggregation. Ridge used $\alpha \in \{0.01, 0.1, 1, 10, 100\}$. Random forests used 400 trees with `max_features` in $\{0.5, 1.0\}$ and `min_samples_leaf` in $\{1, 2\}$. Histogram gradient boosting used 400 iterations, learning rate 0.05, leaf counts in $\{31, 63\}$, and L_2 regularization in $\{0, 1\}$. LightGBM used 600 trees, learning rate 0.05, leaf counts in $\{31, 63\}$, child counts in $\{20, 50\}$, L_2 regularization 1, and row/column subsampling fractions 0.9. A hyperparameter setting and then a learned family were selected independently inside each split by validation MAE; the training-target mean predictor was reported separately and never entered family selection.

II. POST-HOC SCHNET GRAPH-READOUT SENSITIVITY

The prespecified periodic SchNet comparator follows the original energy-oriented construction [1]: its learned atomwise scalar contributions are summed to produce the graph prediction. Defect formation energy is instead localized and non-extensive with respect to supercell atom count, and prior atomistic studies show that graph pooling can materially affect localized or intensive quantum-property prediction [2]. We therefore performed one bounded post-hoc sensitivity analysis after completing the main comparison.

The intervention changed only the graph readout from atomwise addition to mean pooling. It retained all five immutable host-held-out folds, seeds 342–344, the SchNet interaction stack, learned atomic-number input, optimizer, learning rate schedule, host-balanced sampler, target normalization, target-noise stream, gradient clipping, and 150-epoch budget. Predictions were averaged over the three seeds within each fold. The pooled mean-minus-add difference in absolute error was evaluated with 20,000 bootstrap samples clustered by the 44 host materials.

TABLE II. Post-hoc SchNet graph-readout sensitivity under the fixed host-held-out protocol. Each fold prediction averages seeds 342–344. The difference is mean-readout MAE minus additive-readout MAE, in eV. The pooled host-cluster bootstrap 95% interval for this difference is [-7.704, -1.256] eV. The prespecified additive-readout result remains the main comparator.

Partition	n	Add MAE	Mean MAE	Mean − add
Fold 1	2023	3.882	2.179	-1.704
Fold 2	2019	8.199	2.068	-6.132
Fold 3	2069	5.484	2.530	-2.954
Fold 4	2057	5.466	2.468	-2.998
Fold 5	2056	10.474	3.501	-6.973
Pooled OOF	10224	**6.703**	**2.552**	**-4.151**

Mean pooling reduced the host-held-out MAE in all 5/5 folds. Across individual samples, the Spearman association between atom count and absolute error changed from 0.278 for additive pooling to 0.010 for mean pooling. Across the 44 hosts, the corresponding association between median atom count and host MAE changed from 0.312 to 0.218. These descriptive associations are consistent with a size-sensitive additive readout, but atom count covaries with host chemistry and therefore does not identify a standalone causal mechanism. The intervention establishes that readout choice contributes to the extreme additive-SchNet error; it neither replaces the prespecified main comparator nor attributes the remaining DART–SchNet difference to a single component.

[1] K. T. Schütt, H. E. Sauceda, P.-J. Kindermans, A. Tkatchenko, and K.-R. Müller, SchNet – a deep learning architecture for molecules and materials, The Journal of Chemical Physics **148**, 241722 (2018).

[2] D. Buterez, J. P. Janet, S. J. Kiddle, D. Oglic, and P. Liò, Modelling local and general quantum mechanical properties with attention-based pooling, Communications Chemistry **6**, 262 (2023).